

Predictors of Victory in the Indian Premier League

An Exploratory Data Analysis in R

MINI PROJECT – MCA

A PROJECT REPORT

Submitted by

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IN

DATA SCIENCE



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Certificate

This is to certify that the project report entitled "Predictors of Victory in the Indian Premier League: An Exploratory Data Analysis in R" is a bonafide record of work carried out by [Your Name], bearing roll number [Enter number], under my supervision and guidance, in partial fulfilment of the requirements of the course. To the best of my knowledge, the work reported here has not been submitted to any other institution or university for the award of any degree or diploma.

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Declaration

I hereby declare that the project report entitled "Predictors of Victory in the Indian Premier League: An Exploratory Data Analysis in R" submitted by me is an original piece of work carried out using the publicly available IPL dataset and the R programming language. All sources of data, software and ideas used in this report have been duly acknowledged. This report has not been submitted, in part or in full, for any other degree, diploma or course.

Date: 30/10/2026 Signature: [AMAAN HASSAN]

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Abstract

The Indian Premier League (IPL) is the most-watched Twenty20 cricket league in the world. In a format of only 40 overs per match, small advantages can decide a result. This project investigates which factors are associated with winning an IPL match using the programming language R. The study is organised around five research questions: (1) whether winning the toss improves the chance of winning the match; (2) whether it is better to bat first or to chase, and how this varies by season and venue; (3) whether a home-ground advantage exists; (4) which phases of an innings (powerplay, middle overs, death overs) and which batting indicators separate winners from losers; and (5) which predictors remain important when they are considered together in a statistical model.

The analysis uses match-level data, ball-by-ball delivery data, home/away records, career batting summaries and player profiles for the seasons 2008 to 2019. After cleaning and harmonising team names, the report applies descriptive statistics, binomial and proportion tests, Welch t-tests and correlation analysis. Ball-by-ball records are converted into team-innings features such as phase-wise run rate, boundary percentage, dot-ball percentage and wickets lost. Finally, a logistic regression model on standardised between-team differences is fitted and evaluated on a held-out test set, and a random forest is used as a cross-check of variable importance.

The report documents the complete workflow, from data loading to modelling, and the reproducible R Markdown source from which every figure and table is generated. It also discusses the limitations of the approach, notably that in-match indicators explain results rather than forecast them, and proposes future work including team-strength ratings and a true pre-match prediction model.

Keywords: IPL, Twenty20 cricket, exploratory data analysis, toss, home advantage, logistic regression, random forest, R, tidyverse.

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1. Introduction

1.1 Background

Cricket is one of the most followed sports in the world, and since its launch in 2008 the Indian Premier League has become its most commercially successful domestic competition. Each IPL match is a Twenty20 contest in which two teams face a maximum of twenty overs each. Because the match is short, the margin between winning and losing is often very small, and teams invest heavily in analytics to find even marginal advantages.

Analysts, commentators and captains routinely debate a handful of recurring questions. Is the toss really important? Should the captain who wins the toss bat or field? Do teams play better in front of their home crowd? Does a fast powerplay or a strong finish matter most? This project treats these questions as statistical hypotheses and tests them on real match data using a transparent, reproducible workflow in R.

1.2 Motivation

Popular opinion about cricket is full of folklore. A rigorous exploratory data analysis helps separate effects that are supported by the data from those that merely feel true. The project also offers a practical opportunity to apply the complete data-science cycle - importing, cleaning, transforming, visualising, testing and modelling - to a rich, real-world sports dataset.

1.3 Problem Statement

To identify and quantify the factors that are associated with victory in Indian Premier League matches, using match-level and ball-by-ball data from 2008 to 2019, and to determine which of these factors remain important when considered together in a predictive model.

1.4 Objectives

- To clean and harmonise the IPL match, delivery, home/away, batsman and player datasets.
- To test whether winning the toss is associated with winning the match.
- To compare the success of teams batting first and chasing, across seasons and venues.
- To measure home-ground advantage using pooled home and away win rates.
- To engineer team-innings features (powerplay, middle-over and death-over scoring, boundaries, dot balls, wickets) from ball-by-ball data and compare winners with losers.
- To build and evaluate a logistic regression model, and to cross-check variable importance with a random forest.
- To produce a fully reproducible report in R Markdown.

1.5 Research Questions

#	Research question	Method used
RQ1	Does winning the toss improve the chance of winning the match?	Exact binomial test against 50%
RQ2	Is it better to bat first or chase? Does this vary by season and venue?	Binomial test, season and venue summaries
RQ3	Is there a home-ground advantage?	Two-sample test of proportions
RQ4	Which innings phases and indicators separate winners from losers?	Feature engineering, boxplots, Welch t-tests, correlation
RQ5	Which predictors matter together?	Logistic regression with odds ratios, random forest

1.6 Scope of the Study

The study covers IPL seasons 2008 to 2019 as supplied in the project data folder. Only matches with a decisive result are used to study victory. Ball-by-ball modelling is restricted to regulation innings from matches where the Duckworth-Lewis method was not applied. The study is explanatory in nature: it describes which in-match indicators differ between winners and losers rather than attempting to forecast a match before the first ball.

1.7 Organisation of the Report

Chapter 2 reviews the cricket and statistical concepts used. Chapter 3 describes the data. Chapter 4 lists the tools and technologies. Chapter 5 explains the methodology. Chapters 6 to 12 present the analysis in the order of the research questions, followed by the player analysis. Chapter 13 discusses the findings, Chapter 14 states the limitations and Chapter 15 gives the conclusion and future scope.

2. Background and Concepts

2.1 The Twenty20 Format and the IPL

A Twenty20 innings consists of at most twenty overs of six legal deliveries each. The team batting first sets a target, and the team batting second (the chasing side) wins if it passes that target before its overs or wickets run out. Matches are decided in roughly three hours, which magnifies the effect of individual overs. If scores are level at the end of the match, a super over is played to decide the winner, so a tie still yields a winner in the data. Matches that are abandoned without a ball being bowled, or that end without a result, have no winner.

2.2 Phases of an Innings

Phase	Overs	Typical characteristics
Powerplay	1 - 6	Fielding restrictions; batting sides try to score quickly while the field is spread inside the circle.
Middle overs	7 - 15	Consolidation; spinners bowl; teams rebuild or accelerate depending on wickets in hand.
Death overs	16 - 20	Maximum acceleration; boundary hitting; specialist yorker bowlers.

2.3 The Toss

Before the match the captains toss a coin. The winner chooses whether to bat or to field first. Factors such as dew, pitch behaviour and the ability to know the target when chasing have led most captains in recent seasons to choose to field. Whether this choice and the toss itself translate into a statistical advantage is the subject of Research Questions 1 and 2.

2.4 Home Advantage

Home advantage refers to the tendency of teams to perform better at their own ground because of familiarity with conditions, crowd support and reduced travel. In the IPL, a team has a home venue in its franchise city, although some matches in some seasons were played at neutral venues abroad, which dilutes the measurable effect.

2.5 Statistical Concepts Used

- **Binomial test:** tests whether an observed number of successes in n trials is compatible with a hypothesised probability (here 50%). A confidence interval for the true proportion is reported.
- **Test of two proportions:** compares two independent proportions, such as home and away win rates.
- **Welch t-test:** compares the means of two groups (winners and losers) without assuming equal variances.
- **Correlation:** measures the strength and direction of the linear association between two variables, from -1 to +1.
- **Logistic regression:** models the log-odds of a binary outcome as a linear function of predictors. Exponentiated coefficients are odds ratios: values above 1 raise the odds of the outcome.
- **Random forest:** an ensemble of decision trees. Variable importance, measured by the mean decrease in Gini impurity, shows which predictors the trees rely on most.
- **Train/test split:** the model is fitted on 80% of matches and judged on the remaining unseen 20%, with a majority-class baseline for comparison.

3. Dataset Description

3.1 Source

The data come from the publicly available Kaggle Indian Premier League dataset covering the seasons 2008 to 2019, supplied in the data/ folder of the project. Five files are used.

3.2 Files

File	Granularity	Content

matches.csv	One row per match	Teams, season, date, venue, toss winner and decision, winner, win margin by runs or wickets, Duckworth-Lewis flag, player of the match, umpires.
deliveries.csv	One row per ball	Match id, inning, batting and bowling team, over, batsman and bowler, runs (batsman, extras, total), wide and no-ball runs, dismissed player, super-over flag.
teamwise_home_and_away.c sv	One row per team	Home and away matches, wins and win percentages.
most_runs_average_strikerat e.csv	One row per batsman	Career runs, batting average and strike rate.
Players.xlsx	One row per player	Player profile: batting hand, bowling skill, country.

3.3 Key Variables Used

Variable	File	Role in the analysis
season / Season	matches	Converted to an integer year for season-wise summaries.
team1, team2, toss_winner, winner	matches	Team names are standardised; winner defines the outcome.
toss_decision	matches	Bat or field choice of the toss winner.
venue	matches	Ground; used for the venue effect.
win_by_runs, win_by_wickets	matches	Margin of victory; distinguishes bat-first and chasing wins.
dl_applied	matches	Duckworth-Lewis flag; such matches are excluded from ball-by-ball modelling.
over, total_runs, batsman_runs, wide_runs, noball_runs, player_dismissed, is_super_over	deliveries	Raw inputs for phase-wise run rate, boundaries, dot balls and wickets.
home_wins, home_matches, away_wins, away_matches	home/away	Pooled home and away win rates.
total_runs, average, strikerate	batsmen	Quality of leading batsmen.

3.4 Data Quality Considerations

Several issues must be handled before analysis. The season variable is stored as text and must be converted to a year. Dates are stored as day-month-year text. Team names are inconsistent: the Rising
Pune Supergiant franchise appears under two spellings, and Delhi Daredevils were renamed Delhi Capitals. The column umpire3 is almost entirely empty and is not used. Matches without a winner correspond to no-result games and are removed from analyses of victory.

4. Tools and Technologies

4.1 The R Language and RStudio

R is an open-source language and environment designed for statistical computing and graphics. It is widely used in academia and industry because of its extensive package ecosystem, strong visualisation capabilities and support for reproducible research. The project was developed in RStudio, an integrated development environment that provides a script editor, console, plot viewer and one-click report generation through R Markdown.

4.2 R Packages

Package	Purpose in this project

tidyverse (dplyr, tidyr, ggplot2, readr, stringr, purrr)	Data import, wrangling, reshaping, string handling, functional iteration and plotting.
lubridate	Parsing the day-month-year match dates.
readxl	Reading the Players.xlsx workbook.
scales	Formatting axes and inline numbers as percentages.
broom	Converting model output into tidy tables with odds ratios and confidence intervals.
knitr, rmarkdown	Weaving code, output and narrative into one report.
randomForest	Fitting the random forest model and extracting variable importance.

4.3 Project Structure

Item	Role
IPL_EDA_Report.Rmd	Full analysis and report (knit to HTML)
install_packages.R	Installs required packages
README.md	Overview and how to run
R/helpers.R	Team-name cleaning and phase-of-play helpers
data/	Matches, deliveries, home/away, batsmen, players

4.4 Helper Functions and Reproducibility

Three small helper functions are sourced from R/helpers.R: **standardise_team()** harmonises franchise names, **short_team()** produces short labels for plots and **over_phase()** maps an over number to powerplay, middle or death. The report is a single R Markdown document: every number quoted in the text is computed inline from the data, and a fixed random seed (2026) makes the train/test split and the random forest repeatable. Running the project requires three steps: open the folder in RStudio as a project; run install_packages.R once; then open IPL_EDA_Report.Rmd and click Knit.

5. Methodology

The analysis follows the standard data-science workflow: import, tidy, transform, visualise, model and communicate. **5.1 Workflow**

Stage	Activity	Output
1. Import	Read the five data files with readr and readxl.	matches, deliveries, home_away, batsmen, players
2. Clean	Parse dates and seasons, standardise team names, remove no-result matches.	decided (matches with a winner)
3. Derive	Create indicators: toss winner won, bat-first team, bat-first won.	Binary outcome variables
4. Explore	Plot and test toss, batting order, venue, home advantage.	Charts and hypothesis tests
5. Engineer	Aggregate ball-by-ball data into team-innings features by phase.	team_innings table
6. Compare	Boxplots, Welch t-tests and correlations for winners vs losers.	Ranked indicator table
7. Model	Between-team differences, logistic regression, random forest.	Odds ratios, accuracy, importance

5.2 Cleaning and Derived Variables

The text season is reduced to its four-digit year, dates are parsed with lubridate and all team columns in both the match and delivery tables are passed through the standardisation helper. Matches with a missing winner are removed. A tie is decided by a super over, so a winner exists and the match is retained. Three derived indicators are then created for each decided match:

- **toss_win_match_win** - TRUE when the toss winner is also the match winner.
- **bat_first_team** - the toss winner if the decision was to bat, otherwise the other team.
- **bat_first_won** - TRUE when the winner is the team that batted first.

5.3 Ball-by-Ball Feature Engineering

Features are computed only for regulation innings (1 and 2) of decided matches in which Duckworth-Lewis was not applied, so that run rates are comparable. For every ball the analysis records whether it was a legal delivery, the phase of play, whether a wicket fell, whether the batsman scored a four or six, and whether it was a dot ball. These are aggregated to one row per team-innings: overall and phase run rates (runs per over from legal balls), boundary_pct, dot_pct, wickets lost (total, powerplay, death) and extras received.

A chasing side often wins before the death overs are bowled, so its death-over run rate would be missing. In that case the overall run rate is used instead, and missing phase wickets are set to zero. This choice is flagged as a limitation in Chapter 14.

5.4 Hypothesis Tests

For RQ1 and RQ2 an exact binomial test compares the observed proportion of toss winners (and of bat-first teams) who win against the null value of 0.5. For RQ3 a two-sample test of proportions compares pooled home and away win rates. For RQ4 a Welch t-test is run for each of nine indicators, and the table is sorted by p-value. Throughout, a significance level of 5% is used and confidence intervals are preferred to bare p-values.

5.5 Predictive Modelling

The model is built from the perspective of the team that batted first. Each predictor is the difference between the bat-first and chasing teams, which controls for the strength of the opposition within the match, and all differences are standardised so that each odds ratio is expressed per one standard deviation. Matches are split 80/20 into training and test sets. A logistic regression is fitted on the training set; its odds ratios and 95% confidence intervals are plotted, and accuracy on the test set is compared with a majority-class baseline. A random forest of 500 trees is fitted on the same data as a non-parametric cross-check.

6. Data Cleaning and League Overview

6.1 Missing Values

The first step after loading the files is to count missing values in every column of the match table. As expected, umpire3 is largely empty and plays no role in the study. Missing winner entries identify matches with no result.

Quantity to record from the knitted output	Value (fill in)
Rows in matches.csv	
Rows in deliveries.csv	
Columns with missing values	
Matches with no result (removed)	
Decided matches used in the analysis	

6.2 Matches per Season

The number of matches per season shows how the league has grown from its eight-team beginnings and how the number of franchises changed with the arrival and exit of teams. Seasons with more matches carry more weight in pooled statistics, which is why season-wise charts are also shown.

Figure 1 placeholder - insert the plot from R chunk *per-season*. Bar chart with the number of matches labelled above each bar.

Figure 1: Matches played per season

6.3 Team Success

A team's overall win percentage is computed by stacking each decided match twice, once from the viewpoint of team1 and once from team2, and counting wins. Teams with fewer than 30 matches are excluded from the chart to avoid noisy percentages. The dashed line at 50% separates teams that win more often than they lose from those that do not.

Figure 2 placeholder - insert the plot from R chunk *team-success*. Horizontal bars sorted by win percentage with a dashed 50% line.

Figure 2: Overall win percentage by team (teams with 30+ matches)

6.4 Winning Margins

Winning margins are measured in runs when the team batting first wins and in wickets when the chasing team wins, so the two distributions are shown separately. Large run margins indicate dominant batting performances, whereas wicket margins indicate how comfortably a target was chased.

Figure 3 placeholder - insert the plot from R chunk *margins*. Two faceted histograms with free scales.

Figure 3: Distribution of winning margins by runs and by wickets

7. Does the Toss Matter? (RQ1)

7.1 Overall Toss Effect

If the toss had no influence on the outcome, the toss winner would win about half of all decided matches. The report counts the matches in which the toss winner also won and runs an exact binomial test of this proportion against 0.5. The confidence interval for the proportion is the key quantity: if it includes 50%, the toss is not a statistically reliable predictor of victory on its own.

Quantity to record from the knitted output	Value (fill in)
Matches in which toss winner also won	
Decided matches	
Observed proportion	
95% confidence interval	
p-value of the binomial test	

7.2 Toss Decision

The report next compares the proportion of toss winners that won when they chose to bat with the proportion that won when they chose to field. A stacked bar chart then shows how captains' preferences have changed across seasons. In recent seasons captains overwhelmingly prefer to field first, which makes it important to study the batting-order effect separately in Chapter 8.

Figure 4 placeholder - insert the plot from R chunk *toss-decision*. Stacked percentage bars per season, orange for bat and blue for field.

Figure 4: Share of toss winners choosing to bat or field, by season

7.3 Toss Effect over Time

A line chart of the toss winners' win rate in each season, with a dashed line at 50%, shows whether any advantage is stable over time or whether apparent effects are produced by random fluctuation. Since a single season contains only about sixty matches, season-wise percentages vary naturally by several points even when the true probability is exactly one half.

Figure 5 placeholder - insert the plot from R chunk *toss-by-season*. Line with markers across seasons, y-axis limited to 30%-70%.

Figure 5: Toss winners' win rate by season

7.4 Interpretation

A confidence interval containing 0.5 means the data cannot distinguish the toss effect from chance, and an interval entirely above 0.5 would indicate a real if modest advantage. Even a real toss effect of one or two percentage points is small compared with the influence of team strength and in-match performance, which are examined in Chapters 10 and 11.

8. Batting First or Chasing? (RQ2)

8.1 Overall Batting-Order Effect

Using the derived bat-first indicator, the report applies the same exact binomial test to the proportion of matches won by the team batting first. A proportion above 50% would favour setting a target, while a proportion below 50% would favour chasing. Because captains choose the batting order partly in response to conditions, this describes the outcomes of the strategy as actually practised rather than a causal effect of batting order.

Quantity to record from the knitted output	Value (fill in)
Matches won by team batting first	
Bat-first win proportion	
95% confidence interval	
p-value	

8.2 Season Effect

The bat-first win rate is computed for each season and plotted against the 50% line. Persistent movement above or below 50% across consecutive seasons would suggest a structural change in the game, such as the shift towards chasing as the preferred tactic or changes in pitch preparation.

Figure 6 placeholder - insert the plot from R chunk *bat-first*. Green columns per season with dashed 50% reference line.

Figure 6: Win rate of the team batting first, by season

8.3 Venue Effect

Grounds differ in size, pitch and conditions, so the advantage of batting first can differ considerably between them. For every venue with at least 20 matches the report calculates the proportion of bat-first wins and ranks the venues. Venues at the top favour sides that bat first, while those at the bottom favour chasing sides.

Figure 7 placeholder - insert the plot from R chunk *venue*. Sorted horizontal bars with a dashed 50% line.

Figure 7: Bat-first win percentage by venue (20+ matches)

Quantity to record from the knitted output	Value (fill in)
Venue most favourable to batting first	

Venue most favourable to chasing	
Number of venues with 20+ matches	

9. Home-Ground Advantage (RQ3)

9.1 Team-wise Comparison

The home/away file reports the number of matches, wins and win percentages for each franchise at home and away. A grouped bar chart places each team's home and away win percentage side by side.
Teams whose home bar is clearly taller than the away bar benefit from familiarity with their ground.

Figure 8 placeholder - insert the plot from R chunk *home*. Grouped horizontal bars, dark for home and amber for away.

Figure 8: Home versus away win percentage by team

9.2 Pooled Test

Home wins and matches are summed over all teams, as are away wins and matches. A two-sample test of proportions then asks whether the pooled home win rate differs from the pooled away win rate, reported with a confidence interval for the difference.

Quantity to record from the knitted output	Value (fill in)
Pooled home wins / home matches	
Pooled away wins / away matches	
Difference in win rate (home - away)	
95% confidence interval of the difference	
p-value of the proportion test	

9.3 Interpretation and Caveats

A significant positive difference would be evidence of home advantage. However, matches played at neutral venues, such as South Africa in 2009 and the United Arab Emirates in 2014, do not have a genuine home team, and where labelled as home or away they dilute the measured effect towards zero. Because each completed match produces one win and one loss, the pooled rates also depend strongly on the fixture schedule. The comparison should be read as a descriptive indicator of the size of the effect and not as a precise estimate.

10. Ball-by-Ball Features: Winners versus Losers (RQ4)

10.1 The Team-Innings Table

After feature engineering, the analysis has one row per team-innings containing the run rates for each phase, boundary percentage, dot-ball percentage, wickets lost and an indicator of whether the team won, plus indicators for batting first and winning the toss.

Quantity to record from the knitted output	Value (fill in)
Number of team-innings in the table	
Number of matches represented	
Average overall run rate (runs per over)	

10.2 Distributions for Winners and Losers

Boxplots for each of the nine indicators compare winning and losing innings. A clear vertical shift of the winner box relative to the loser box indicates an indicator that separates the two groups. Because the winner of each match is also the loser's opponent, the boxplots naturally show large differences in overall run rate and wickets.

Figure 9 placeholder - insert the plot from R chunk *winners-losers*. Faceted boxplots with green for winners and red for losers.

Figure 9: Team-innings indicators for winners and losers

10.3 Welch t-tests

For every indicator a Welch t-test compares the means of winners and losers. The table is sorted from the strongest to the weakest evidence, ranking the indicators by how well they separate the two groups one at a time.

Feature	Mean (loser)	Mean (winner)	Difference	p-value
powerplay_rr				
middle_rr				
death_rr				
overall_rr				
boundary_pct				
dot_pct				
wickets				
powerplay_wkts				
death_wkts				

10.4 Correlation between Indicators

The correlation matrix includes the win indicator together with all nine features. It shows the correlation of each feature with winning, and the correlations between features, which warn of multicollinearity. For instance, overall run rate is mechanically related to boundary percentage and dot-ball percentage, so these should not be read as independent effects.

Figure 10 placeholder - insert the plot from R chunk *corr*. Heat map with correlation coefficients in each tile.

Figure 10: Correlation matrix of winning and team-innings indicators

10.5 Batting First versus Chasing

Finally, average phase run rates and the win rate are compared between teams batting first and chasing. This describes how scoring patterns differ between the two innings, partly because a chasing team adapts its tempo to the target, and partly because chasing innings are truncated when the target is reached.

Quantity to record from the knitted output	Value (fill in)
Mean powerplay run rate: batting first / chasing	
Mean middle-over run rate: batting first / chasing	
Mean death-over run rate: batting first / chasing	

11. Predictive Modelling (RQ5)

11.1 Model Set-up

Univariate comparisons cannot say which indicators matter once the others are held constant. The report therefore models the outcome from the perspective of the team batting first. The response is bat_first_won, and the predictors are whether the bat-first team won the toss together with standardised differences (bat-first minus chasing) in powerplay, middle and death-over run rates, boundary percentage, dot-ball percentage and wickets lost.

An important caveat: wickets lost by a chasing team are mechanically linked to the result, since a chase succeeds before the wickets run out. The model is therefore explanatory and not a pre-match forecast. A genuine pre-match model would use only information available before the first ball, such as the toss, the venue and team strength.

11.2 Logistic Regression

The data are divided at random into a training set (80%) and a test set (20%) with a fixed seed, and a binomial generalised linear model is fitted on the training set. The exponentiated coefficients give odds ratios with 95% confidence intervals. An odds ratio above 1 means that a one-standard-deviation advantage for the bat-first team on that indicator raises the odds that it wins; below 1 lowers them. An interval that crosses 1 means the effect is not distinguishable from zero after adjusting for the other predictors.

Term	Odds ratio	95% CI low	95% CI high
toss_won_batfirst			
d_powerplay_rr			
d_middle_rr			
d_death_rr			
d_boundary_pct			
d_dot_pct			
d_wickets			

Figure 11 placeholder - insert the plot from R chunk *odds-ratios*. Point-range plot on a log scale with a dashed line at 1.

Figure 11: Odds ratios for the bat-first team winning (95% CI)

11.3 Performance on Unseen Matches

Predicted probabilities are calculated for the test matches and converted to classes with a cut-off of 0.5. Accuracy is compared with a majority-class baseline, the accuracy obtained by always predicting the more common result. A model is only useful if it clearly outperforms this baseline.

Quantity to record from the knitted output	Value (fill in)
Number of training matches	
Number of test matches	
Test accuracy	
Majority-class baseline	
True/false positives and negatives	

11.4 Random Forest Cross-Check

The random forest, with 500 trees, does not assume linearity or additivity. Its importance measure, the mean decrease in Gini, ranks the predictors by how much they improve the purity of the splits. If the forest and the logistic regression agree on the leading predictors, confidence in the findings is strengthened; disagreement may signal non-linear effects or interactions.

Figure 12 placeholder - insert the plot from R chunk *rf*. Horizontal bars sorted by importance.

Figure 12: Random forest variable importance (mean decrease in Gini)

12. Players and Batting Quality

12.1 Player of the Match

Counting Player-of-the-Match awards identifies the players who most often produce match-winning performances. The report charts the twelve players with the most awards. This is a descriptive view influenced by career length, since players with more seasons have more opportunities to win awards.

Figure 13 placeholder - insert the plot from R chunk *pom*. Horizontal bars of the top twelve award winners.

Figure 13: Players with the most Player-of-the-Match awards

12.2 Batting Average versus Strike Rate

Batting average rewards consistency while strike rate rewards speed, and in Twenty20 cricket the best batsmen combine both. A scatter plot of average against strike rate for batsmen with at least 1,500 IPL runs, sized by total runs, shows who sits in the high-average, high-strike-rate corner.

Figure 14 placeholder - insert the plot from R chunk *batsmen*. Bubble plot sized by career runs.

Figure 14: Batting average versus strike rate (1,500+ IPL runs)

12.3 Player Profiles

The player workbook allows a count of batting hands (left- and right-handed batsmen), giving context about the composition of IPL squads. It is a descriptive supplement and is not used in the models.

13. Discussion

The analysis is structured so that each research question has a specific statistical answer which is produced automatically whenever the report is knitted. The table below explains how those answers should be read together.

Research question	Evidence in the report	How to read it
RQ1 Toss	Binomial test, confidence interval, season line chart	Interval including 50% means no reliable toss effect; look at season variability before concluding.
RQ2 Batting order	Bat-first win rate overall, by season and by venue	Distinguish a general tendency from venue-specific conditions; use only venues with enough matches.
RQ3 Home advantage	Pooled home vs away test, team-wise bars	A positive significant difference supports home advantage; neutral venues weaken it.
RQ4 Phases	t-test table, boxplots, correlation heat map	Indicators with the smallest p-values separate winners from losers; beware of overlap between features.
RQ5 Model	Odds ratios, accuracy vs baseline, forest importance	Predictors whose intervals exclude 1 stay important after adjustment; agreement with the forest adds confidence.

A recurring theme in sports analytics is that the observable events of a match are largely a consequence of the quality of the two teams. For that reason the report separates the question of what differs between winners and losers (Chapter 10) from the question of what remains important once everything is considered together (Chapter 11), and it warns that in-match features cannot be used to forecast results before the match begins.

14. Limitations

- **Time span.** The data cover only the seasons 2008 to 2019; later rule changes, team composition and tactics are not represented.

- **Team strength.** Squad quality is not modelled directly, so differences due to player quality may be attributed to other indicators.
- **Truncated chases.** Chasing innings end when the target is reached, so their phase run rates, death-over rates and wickets are affected by the result itself. Replacing missing phase values with the overall run rate is a pragmatic approximation.
- **Excluded matches.** Duckworth-Lewis and no-result matches are omitted from the ball-by-ball analysis, which slightly reduces the sample.
- **Explanatory, not predictive.** Predictors are measured during the match, so they cannot be used before the first ball.
- **Neutral venues.** Matches played outside India dilute the home-advantage measurement.
- **Multiple comparisons.** Many indicators and venues are tested; a few small p-values may arise by chance, so effect sizes and confidence intervals deserve more attention.
- **Small samples.** Season-wise and venue-wise percentages rest on a limited number of matches and vary naturally.

15. Conclusion and Future Scope

15.1 Conclusion

This project built a complete and reproducible exploratory analysis of the factors associated with victory in the Indian Premier League. Starting from raw match and ball-by-ball files, it cleaned and harmonised the data, engineered phase-wise team-innings features and addressed five research questions on the toss, batting order, venues, home advantage, in-match indicators and multivariate prediction using binomial and proportion tests, Welch t-tests, logistic regression and a random forest. The work demonstrates how a set of simple, transparent statistical tools can turn popular beliefs about cricket into testable hypotheses, and how a single R Markdown document can keep the code, the numerical results and the written interpretation synchronised.

15.2 Future Scope

- Add team-strength ratings such as Elo or Bayesian ratings to account for squad quality.
- Add player-level features such as the batting average and strike rate of the line-up, and bowler quality.
- Build a true pre-match model that uses only the toss, venue, recent form and team strength, and evaluate it with time-based validation.
- Extend the dataset to the seasons after 2019 and compare tactical trends over time.
- Develop a live win-probability model that updates after every over.
- Publish the findings as an interactive Shiny dashboard.

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Appendix: Checklist for Completing the Report

The report text and methodology are complete. Because the data files were not part of the upload, the numerical results were left to be filled in from the knitted output rather than invented. To finish the report:

- Knit IPL_EDA_Report.Rmd with the data folder in place.
- Replace each figure placeholder with the matching plot (export with ggsave() or screenshot the HTML output).
- Fill in the "Quantity to record" tables and the t-test and odds-ratio tables from the knitted output.
- Write one sentence after each table stating the finding (for example whether the toss confidence interval includes 50%).
- Enter your name, roll number, institution and guide on the cover, certificate and declaration pages.

Output:

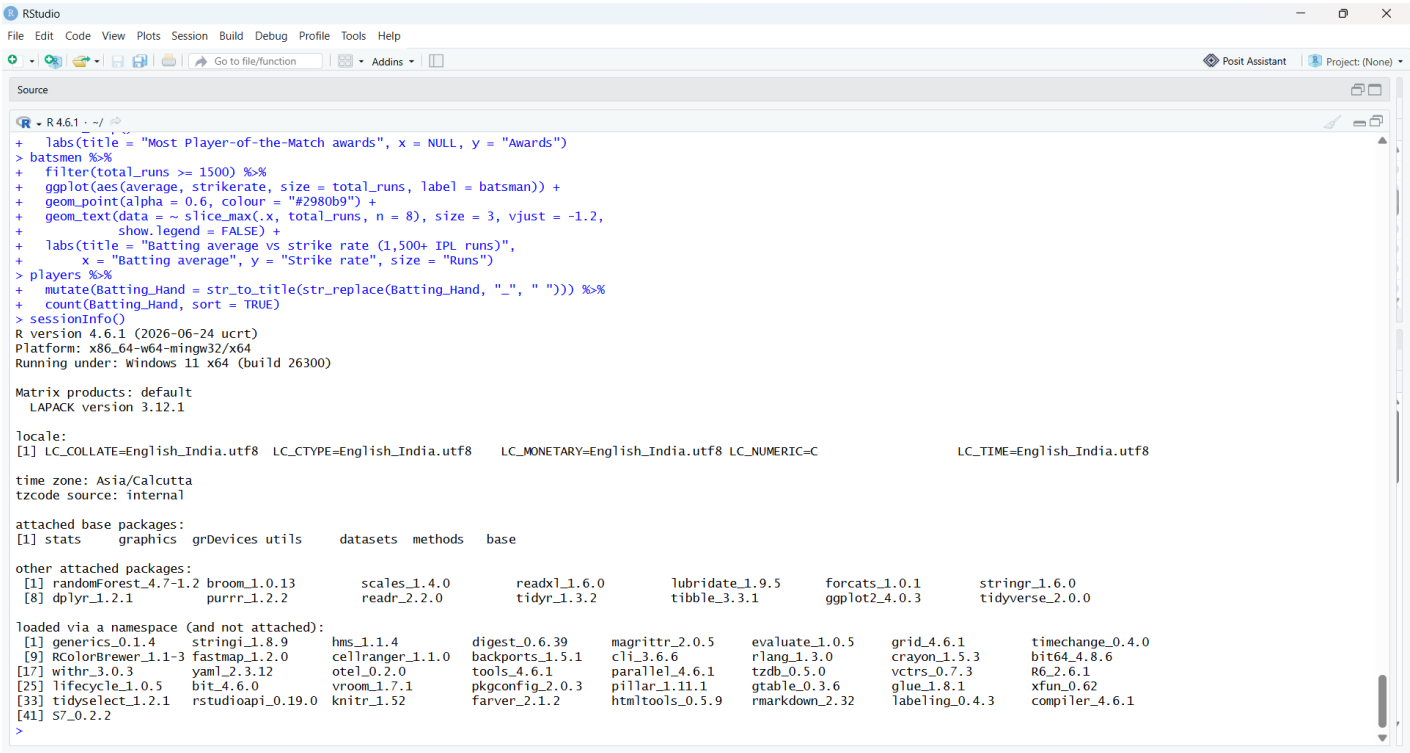


Fig no. 1

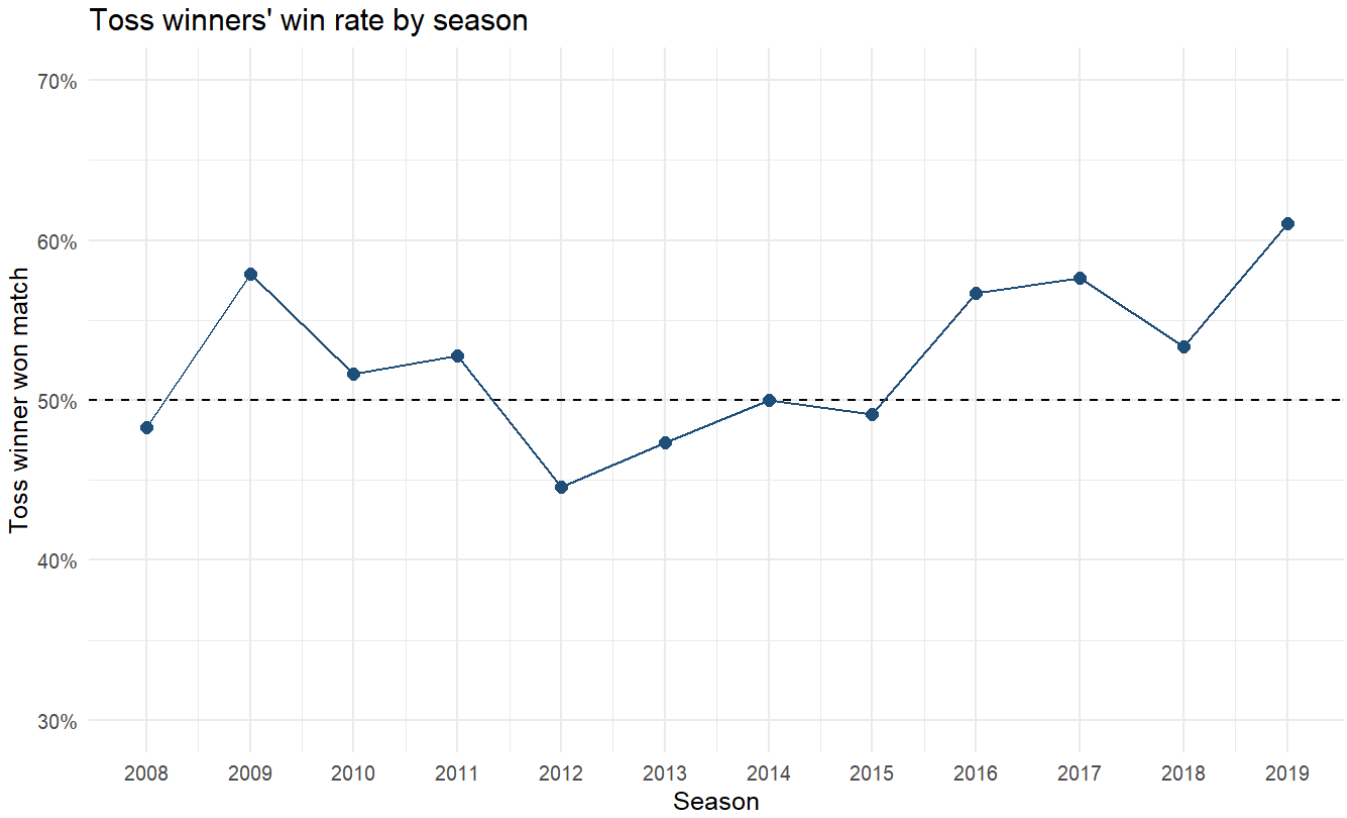


Fig no. 2

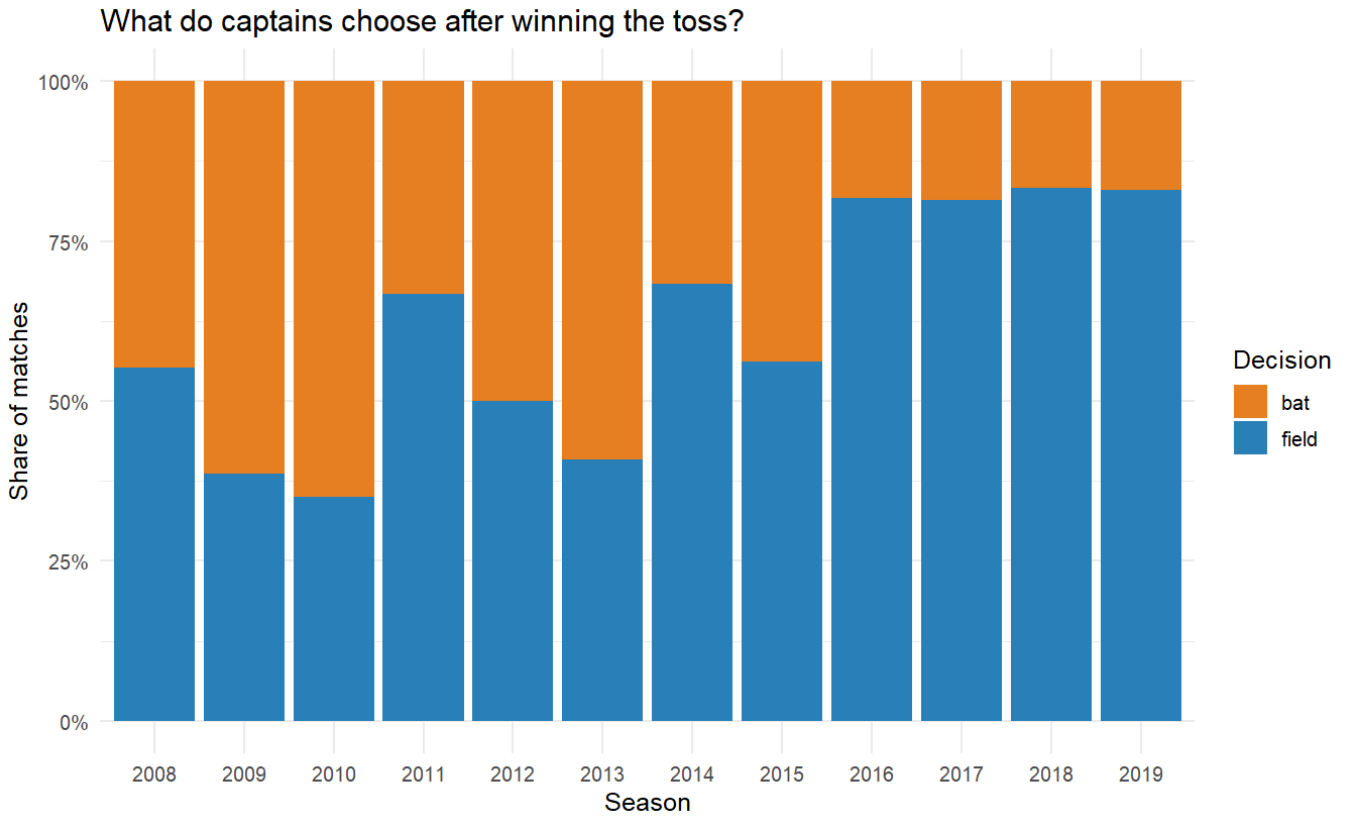


Fig no. 3

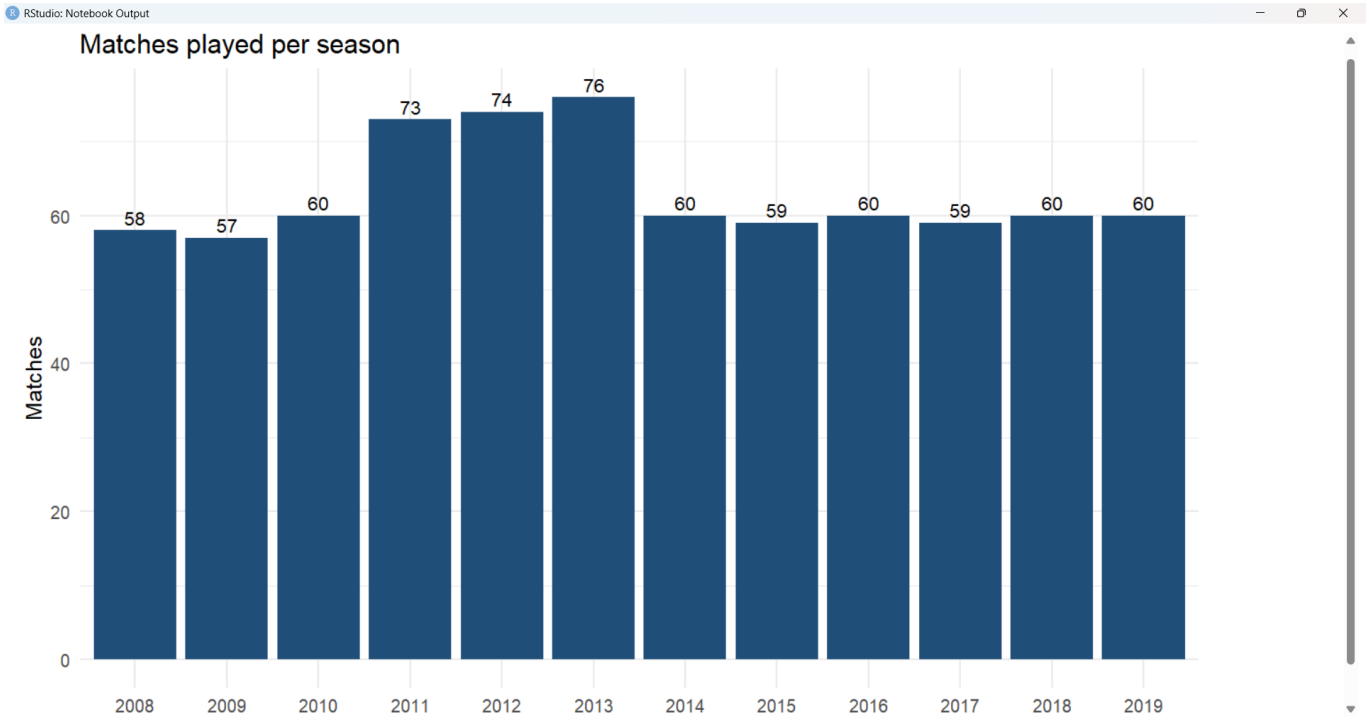


Fig no.4

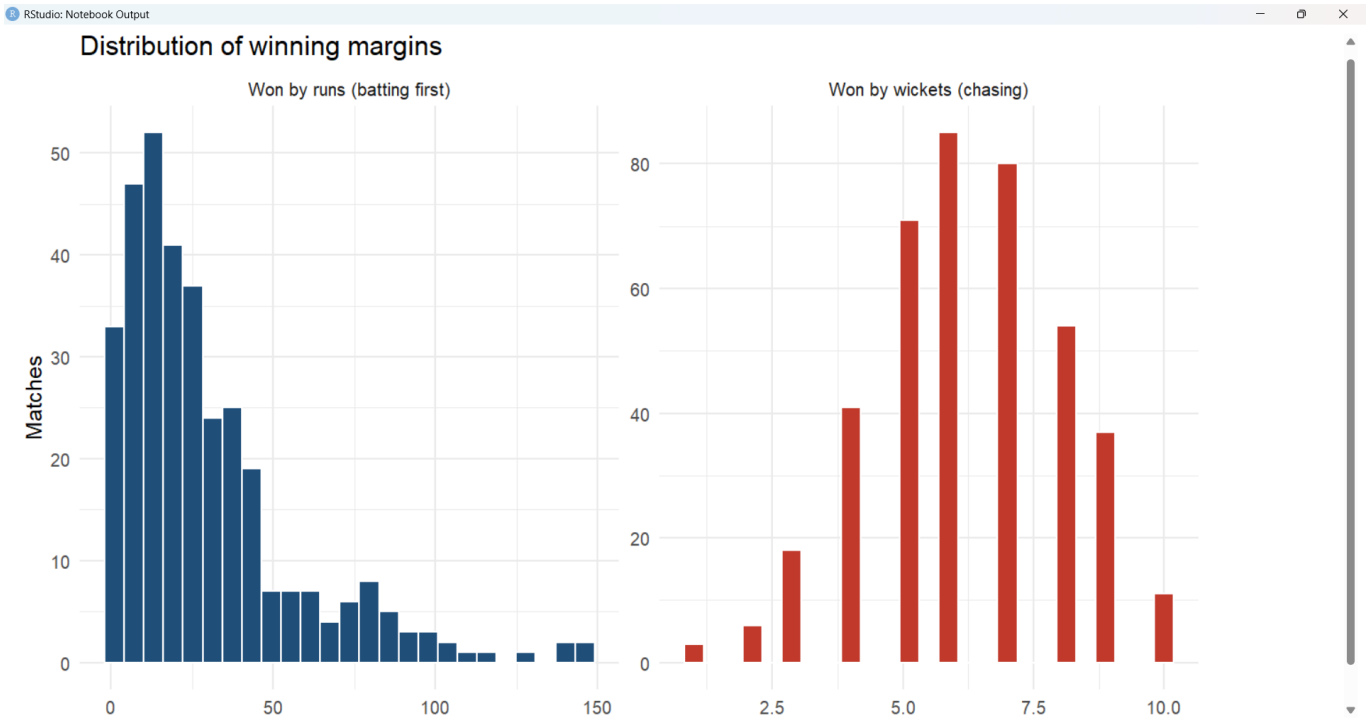


Fig no. 5